The Hidden Costs of Chemical Accidents: Evidence from Downwind Schools

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Abstract

About two-thirds of U.S. public school students attend school within 10 km of a facility that uses or stores extremely hazardous chemicals. I estimate the impact of accidental releases on student learning. Linking airborne accidents with off-site impacts in 2009–2018 to test scores in nearby schools, I exploit quasi-exogenous variation in wind direction on the accident day, comparing downwind schools with upwind and crosswind schools. Math scores fall by 0.040 school-level standard deviations on the first test after an accident and are 0.056 SD lower on average over the first five test years. Reading effects are smaller and imprecise. Losses are concentrated in accidents with the highest toxicity-weighted release rates, and the persistent math decline falls mainly on non-white students. The results suggest that, by omitting learning losses, regulatory analyses understate the benefits of chemical accident prevention.

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1 Introduction

About 131 million people live within three miles of a facility regulated under EPA's Risk Management Program (RMP), which covers facilities that hold extremely hazardous chemicals above threshold quantities (U.S. EPA, 2024b).[1] When an accident at one of these facilities releases toxic substances beyond the fence line, it poses an acute threat to the surrounding community, and schools are often part of that community: about two-thirds of U.S. public school students attend a school within 10 km of at least one RMP facility.[2] Yet we know little about whether a single accidental release affects children's human capital, and these effects are absent from the regulatory accounting of accident prevention.

This paper estimates the human capital cost of chemical accidents. I link RMP accident records from 2009 to 2018, obtained through the Data Liberation Project's FOIA requests, to school locations from the Department of Education's Common Core of Data and to school-level standardized test results for grades 3–8 from EDFacts for school years 2009–10 to 2018–19. For airborne accidents with off-site impacts, I exploit quasi-exogenous variation in wind direction on the accident day, a source of variation widely used to study air pollution (Deryugina et al., 2019; Chung et al., n.d.). Among schools within 10 km of the reporting facility, treated schools are those downwind on the accident day and control schools are those upwind or crosswind, and I estimate an event-study difference-in-differences (DID) model.

Because treated and control schools lie near the same facility and at similar distances from it, the comparison holds the facility's presence fixed and isolates the effect of the release itself. Because accidents occur at different times, I show that the results are similar with heterogeneity-robust staggered DID estimators (Sun and Abraham, 2021; Wing et al., 2024).

I find that being downwind of an airborne accident with off-site impacts lowers math achievement by 0.040 school-level standard deviations (SD) on the first test after the accident. Averaged over the first five test years, math scores are 0.056 SD lower. Reading scores show a smaller and only marginally significant immediate drop. In student-level terms, the immediate

[1] EPA presented these numbers in its 2024 report; of these residents, approximately 20 million identify as Black or African American, 32 million as Hispanic or Latino, and 44 million have incomes at or below twice the poverty level (U.S. EPA, 2024b).

[2] Author's calculation, matching all RMP-registered facilities to public school locations and enrollment from the Common Core of Data.

effect is roughly 0.017 SD,[3] comparable to the effect of a TRI facility opening within one mile of a school (Persico and Venator, 2021) and about half that of attending a school downwind of a highway (Heissel et al., 2022), even though the exposure is a single acute release. This immediate effect is robust to heterogeneity-robust estimators and alternative comparison groups. It is not explained by schools' long-run position downwind of the facility or by changes in who takes the test: the number of students tested, the non-white share of test-takers, and enrollment do not change after an accident.

Three pieces of evidence suggest that the effects operate through exposure to the release. First, using the toxic endpoints EPA prescribes for off-site consequence analysis (40 CFR 68, Appendix A), I construct each accident's toxicity-weighted release rate. The effects are driven by middle- and high-toxicity releases, while low-toxicity and purely flammable releases show no decline. Second, accidents whose harm stayed on site show no effect on downwind schools. Third, lost instructional time is unlikely to explain the results: the immediate effect is as large for accidents outside school hours, including summer accidents.

The effects are larger in grades 6–8 (-0.055 SD on impact) than in grades 3–5 (-0.026 SD), although the difference is not statistically significant. Female and male students show nearly identical effects. Suggestively, the persistent math decline falls mainly on non-white students: within the same schools, White students' scores return to baseline after the first year, while non-white students' scores remain lower.

First, this paper contributes to understanding the external costs of industrial chemical accidents. Accidents with off-site impacts lower nearby home values by 2–3 percent for at least 15 years (Guignet et al., 2023b,a), and chemical spills can harm health: after the 2014 Elk River spill contaminated drinking water in West Virginia, newborns' Apgar scores fell (Guilfoos et al., 2018). To my knowledge, this paper provides the first estimates of the effect of chemical accidents on children's human capital. Because early test scores predict later educational attainment and earnings (Chetty et al., 2011, 2014), these costs may persist into adulthood. They also fall disproportionately on disadvantaged communities, which are overrepresented near RMP facilities and especially near facilities that have accidents (Guignet et al., 2024).

Second, this paper contributes to the literature on environmental exposures and human

[3] Converting the share proficient with an inverse-normal transformation; scaling by the square root of typical intraclass correlations (Hedges and Hedberg, 2007, 2013) gives 0.016–0.020 SD.

capital. A large literature documents that childhood environmental exposures harm human capital, lowering test scores and raising behavioral incidents, grade repetition, and cognitive disabilities, with consequences for later earnings (Rau et al., 2015; Persico et al., 2020). Most of this evidence concerns chronic or long-term exposure: proximity to a newly opened Toxic Release Inventory facility (Persico and Venator, 2021), early-life exposure to airborne toxic releases (Jacqz, 2022), and coal-fired power production near schools (Duque and Gilraine, 2022). Studies of short-run variation in ambient air quality also show that pollution on exam days lowers scores and later earnings (Ebenstein et al., 2016), that volcanic emissions lower test scores, especially among disadvantaged students (Inafuku et al., 2026), and that daily particulate pollution increases absences and disciplinary referrals (Chung et al., n.d.). I study a single, accidental, high-concentration release. Ambient pollutants such as $PM_{2.5}$ are regulated over continuous exposure, while RMP substances are benchmarked against concentrations harmful after exposures of about one hour. Despite being a one-time event, this exposure lowers math achievement by about 0.017 student-level SD in the first test after the accident, comparable to the effect of a nearby TRI facility opening (Persico and Venator, 2021).

My results show that accidents with off-site impacts impose a human capital cost on downwind schoolchildren. To the extent that accident prevention requirements reduce such accidents, their benefits are larger than the regulatory impact analysis recognizes, since it does not account for effects on children's learning (U.S. EPA, 2024b). Because non-white students are overrepresented near facilities where accidents occur (Guignet et al., 2024), and because the effects I estimate are, if anything, larger for these students, chemical accidents may widen existing gaps in achievement and in later earnings.

These findings are directly relevant to current policy. In 2024, EPA amended the RMP to strengthen accident prevention (U.S. EPA, 2024a); in February 2026, it proposed rescinding or scaling back major parts of those amendments (U.S. EPA, 2026a). The regulatory impact analysis of the 2024 rule valued off-site accident damages at about $42 million per year, almost all of it property damage; hospitalizations, evacuations, and sheltering in place together account for less than $7 million (U.S. EPA, 2024c). Effects on children's learning are not included. The results imply that off-site accident damages are larger than previously thought and that human capital costs should be included in cost-benefit analysis.

2 Data

I construct school-level test scores for downwind and non-downwind schools for school years 2009–19 by linking each accident from 2009 to 2018 to the wind direction and to schools within 10 km, and then linking all test scores.

2.1 Chemical Accidents

Under the EPA's Risk Management Program (RMP), any facility that holds more than a threshold quantity of a regulated toxic or flammable substance in a process must file a risk management plan (U.S. EPA, n.d.a). Each plan reports the facility's latitude and longitude, its hazard assessment and prevention program, and a five-year history of accidental releases from covered processes. For each accident, the facility reports the date, start time, and duration of the release; the chemicals released (e.g., toxic or flammable) and their quantities; the release event (e.g., gas release vs. liquid spill); off-site and on-site impacts; the industry code of the process; and weather conditions. I use the accidents and submissions files of the RMP records released to the Data Liberation Project under the Freedom of Information Act (Data Liberation Project, 2026).

I consider all accidents with off-site impacts, from all sectors, between 2009 and 2018 in the conterminous U.S.[4] Known off-site impacts include deaths, injuries, evacuations, sheltering in place, property damage, or environmental damage in the area surrounding the facility (U.S. EPA, n.d.a, § 68.42). I exclude accidents whose only off-site impact is environmental damage, as no direct human impact is expected. Because a gas release, fire, or explosion puts the chemical, or its combustion products, directly into the air, the wind on the accident day largely determines which nearby schools are exposed. My main sample therefore restricts these accidents to airborne releases, defined as accidents that involve at least one of these three release events (Appendix B.1). Accidents whose only release events are liquid spills or runaway reactions, where vapor forms more slowly and is more likely to stay on site, are included in a robustness check.

Each RMP accident report lists the chemicals released, the quantity of each, and the

[4] These data are self-reported, and only accidents with known or reported impacts are reported, so the number of accidents may be undercounted.

release duration. Because accidents release different chemicals, in different quantities, and over different durations, comparing them requires a common measure of hazard. Following the off-site consequence analysis in EPA's Chemical Accident Prevention Provisions (U.S. EPA, n.d.b, 2009b), I construct a toxicity-weighted release rate for each accident. This analysis characterizes a toxic release by its release rate, the quantity released divided by the release duration, and by the chemical's toxic endpoint, the maximum airborne concentration at which nearly all individuals could be exposed for up to one hour without serious health effects (40 CFR Part 68, Appendix A). Because most toxic accidents in the sample involve ammonia, I express all releases in ammonia-equivalent terms:

$$R_a = \sum_{c \in \mathcal{T}_a} \frac{q_{ac}}{\max\left(h_a, \frac{1}{6}\right)} \cdot \frac{E_{\mathrm{NH_3}}}{E_c} \tag{1}$$

where a indexes accidents and c chemicals; $\mathcal{T}_a$ is the set of regulated toxic substances released in accident a; q_{ac} is the quantity of chemical c released (pounds); h_a is the release duration (hours), floored at 10 minutes, EPA's worst-case release duration for toxic gases; and E_c is the toxic endpoint of chemical c (mg/L). The factor $E_{\mathrm{NH_3}}/E_c$ converts each chemical's pounds into ammonia-equivalent pounds, where $E_{\mathrm{NH_3}} = 0.14$ mg/L is the endpoint of ammonia, the most common chemical in the sample. R_a is the hourly release of ammonia that would be equally hazardous; chlorine, for example, receives a weight of 16.[5] I divide the toxic accidents into low, middle, and high groups. [6]

2.2 Wind

The RMP includes self-reported wind direction (16 compass points) and wind speed at the time of the accident, but these fields are missing for 39% of accidents in the sample.[7] I take wind data from the ERA5 reanalysis (Hersbach et al., 2020, 2023), accessed through the Open-Meteo historical weather API (Zippenfenig, 2023), which provides hourly wind speed and direction

[5] Quantities and durations are self-reported by facilities and are measured with error. EPA does not define thresholds for R_a, so I group accidents by terciles of the identifying toxic accidents.

[6] Flammable substances have no toxic endpoint, so accidents that release only flammable substances form a separate group.

[7] The RMP also does not report how facilities determine wind direction, which may be estimated visually rather than measured with an instrument.

at 10 meters at a resolution of about 31 km. For each accident, I use the hourly wind series at the land grid cell nearest the facility for the accident's local calendar day.[8] Because release timing and duration are reported only approximately, I define each accident's wind direction from the daily vector-mean wind, the average of the 24 hourly wind vectors.

Wind direction follows the meteorological convention: it records the direction from which the wind blows, so a release travels toward the opposite bearing. For example, wind from the east carries a release west. For each school within 10 km of the facility, I compute the angle $\Delta_{ia} \in [0°, 180°]$ between the facility-to-school bearing and the downwind bearing. A school is downwind if it falls within a 90° cone centered on the downwind bearing ($\Delta_{ia} \leq 45°$), upwind if it falls within the opposite 90° cone ($\Delta_{ia} \geq 135°$), and crosswind otherwise.

2.3 School Locations and Characteristics

School locations and characteristics come from the National Center for Education Statistics' Common Core of Data (CCD) (U.S. Department of Education, National Center for Education Statistics, n.d.), the primary database on public elementary and secondary schools, accessed through the Urban Institute's Education Data Portal (Urban Institute, 2026). I use the school directory for every public school in the conterminous United States in each school year from 2009–10 to 2018–19, which gives a school's coordinates, operating status, grade span, locale, charter and virtual status, enrollment, and number of students eligible for free or reduced-price lunch.

The analysis sample consists of schools within 10 km of an accident facility that are open, operate in person, and have a location that can be determined with confidence.[9]

2.4 Test Scores

Test scores come from the U.S. Department of Education's EDFacts public files on state assessments in mathematics and reading/language arts (RLA) for grades 3–8, school years 2009–10 to 2018–19, which I obtain from the Education Data Center's archive of the EDFacts

[8] Open-Meteo matches each facility to an ERA5 grid cell using its default land-cell selection.

[9] Because NCES re-geocodes schools over time, I compare each school's coordinates across years and assign it a single location. Schools whose coordinates and street address both change are excluded, because a relocation cannot be distinguished from an address correction. See Appendix B.

public files (U.S. Department of Education, n.d.).[10] For each school, subject, grade, and year, EDFacts reports the number of students with a valid score and the percentage scoring at or above the state's proficiency level. To protect student privacy, the percentage proficient is published as a range whose width shrinks as the number of students tested grows, and it is suppressed when 1–5 students are tested (Table A3). Suppressed cells have no score and are dropped; they make up 2.1 percent of the cells of the estimation-sample schools but only 0.05 percent of students tested, the regression weight. Among the remaining cells, ranges of 5 percentage points or narrower cover about 63 percent of cells and 85 percent of students tested. Because the number of proficient students must be a whole number, I narrow each range to the rates consistent with an integer count and use the midpoint of the narrowed range, p_{ijgt}, for school i, subject j, grade g, and year t (Appendix B.2).

Because states use different tests and proficiency standards, and change them over time, proficiency rates cannot be compared across states or years. Following Jacqz (2022), I standardize them within each state, subject, grade, and year:[11]

$$z_{ijgt} = \frac{p_{ijgt} - \mu_{Sjgt}}{\sigma_{Sjgt}}, \tag{2}$$

where μ_{Sjgt} and σ_{Sjgt} are the mean and standard deviation of p_{ijgt} in state S, computed over a balanced panel of the state's schools: those with a score in every year in which the state tests that subject and grade. I standardize using all EDFacts schools in the state and only then restrict the sample to schools within 10 km of an accident, so z_{ijgt} measures a school's position in its state's distribution. Estimates are therefore in school-level standard deviations; Section 4 converts them into percentage points proficient and student-level standard deviations. [12]

Most states test in the spring, but seven states tested in the fall through 2013–14.[13] Because

[10] The school-level mathematics file for 2009–10 was retrieved from the Internet Archive's copy of the original.

[11] Jacqz (2022) uses restricted-use EDFacts files, which include proficiency rates that are suppressed or rounded in the public files. With the public files, I first recover an interval for each school's rate and use its midpoint (Appendix B.2).

[12] A year in which fewer than half of the state's usual schools report a score is left out when the balanced panel is defined and receives no standardized score. Cells with fewer than 30 balanced schools also receive no standardized score.

[13] Maine, Michigan, New Hampshire, North Dakota, Rhode Island, Vermont, and Wisconsin tested in the fall through 2013–14 (U.S. Department of Education, 2016b) and moved to spring testing in 2014–15 (U.S. Department of Education, 2016a).

a fall test mainly reflects learning in the previous school year, I use each state's testing calendar to date the first test after each accident.

3 Empirical Specification

Schools near industrial facilities that handle hazardous chemicals might differ from schools farther away. I restrict the sample to schools within 10 km of an RMP facility that had an airborne accident with off-site impacts. Within this sample, the wind on the accident day determines which schools the release reaches. Because the accident data record the date of each accident, I can measure the wind direction on each accident day and classify each nearby school as downwind or not.

Community exposure is relevant when an accident with off-site impacts reaches over the facility fence line, and wind carries airborne releases most directly (Section 2). The main specification uses airborne accidents with off-site impacts, for which the exposure pathway from the facility to the school is complete.[14] All schools in the sample, treated and control, lie within 10 km of such a facility, so the design compares schools in the same proximity band that differ in whether the wind carried a release toward them. In the main specification, a treated school is compared with never-downwind schools in the same state, subject, grade, and year; Section 5 restricts the comparison to schools around the same facility.

For each accident a, I measure the direction toward which the wind is blowing (the downwind bearing) and the bearing from the facility to each school i, so both bearings point away from the facility. Let $\Delta_{ia} \in [0°, 180°]$ be the angle between these two bearings. School i is downwind of accident a if $\Delta_{ia} \leq 45°$, crosswind if $45° < \Delta_{ia} \leq 135°$, and upwind if $\Delta_{ia} > 135°$. A school is treated from its first downwind accident onward; schools that are never downwind of an accident in the sample serve as controls. Figure 1 illustrates how schools around an accident are classified. The design follows studies that use quasi-exogenous variation in the wind direction from a fixed pollution source (Heissel et al., 2022; Duque and Gilraine, 2022) or that use wind direction as an instrument for pollution (Deryugina et al., 2019). The sample includes 283 airborne accidents with off-site impacts at 253 facilities, and the estimation

[14] The broader set of accidents with off-site impacts also includes spills, which may have both liquid and vaporization exposure pathways. See Section 5 for alternative specifications.

sample has 1,180 treated schools and 3,529 control schools.

For a treated school, T_i is the first test year after its first downwind accident. Most states test in the spring, and I use April 1 as the test date: an accident in month m of calendar year y is assigned to spring test year $y + \mathbf{1}\{m \geq 4\}$. An accident between August and March therefore affects the current school year's test scores, while an accident between April and July affects the next school year's test scores.[15]

I estimate the dynamic effects of being downwind of an airborne chemical accident with off-site impacts with the following two-way fixed effects (TWFE) event-study regression:

$$Y_{ijgt} \;=\; \sum_{\substack{k=-4 \\ k\neq -1}}^{4} \beta_k \, D_{it}^{k} \;+\; \alpha_{ijg} \;+\; \gamma_{S(i)jgt} \;+\; \varepsilon_{ijgt}, \tag{3}$$

where Y_{ijgt} is the standardized proficiency rate z_{ijgt} of school i in subject j, grade g, and year t, standardized within state × subject × grade × year so that units are school-level standard deviations; α_{ijg} is a school × subject × grade fixed effect; and $\gamma_{S(i)jgt}$ is a state × subject × grade × year fixed effect, where $S(i)$ is the state of school i. For a treated school with first exposure year T_i, let $\tilde{k}_{it} = \max\{-4, \min\{4, t - T_i\}\}$ be event time, binned at the endpoints, and let $D_{it}^{k} = \mathbf{1}\{i \text{ treated}\} \cdot \mathbf{1}\{\tilde{k}_{it} = k\}$, so that $D_{it}^{k} = 0$ for control schools. The year before first exposure ($k = -1$) is the omitted category, so β_k measures the difference between downwind and control schools k years after first exposure, relative to that year. I summarize the effect by the immediate effect β_0 and by the equally weighted average of $\beta_0, \ldots, \beta_4$, where β_4 includes all later years and β_{-4} all earlier years. The regression is weighted by the number of students with valid scores in each school-subject-grade cell. I cluster standard errors by facility group, the unit that generates treatment: each cluster is a facility that had an accident in the sample, and facilities whose 10 km circles share a school are merged into one cluster following Duque and Gilraine (2022), who cluster at the level of the power plant that generates the pollution variation.

The identifying assumption is that, in the absence of an accident, downwind and non-downwind schools near accident facilities in the same state would have followed similar trends.

[15] According to the EDFacts documentation, seven states (ME, MI, NH, ND, RI, VT, and WI) tested in the fall from 2009–10 through 2013–14. For these state-years, I use October 1 of the preceding calendar year as the test date. Appendix B.2 explains the details.

Accidents are unplanned, so I expect no anticipation effects. Several features of the setting could violate this assumption. First, each location has a prevailing wind, so schools on a facility's usual downwind side might be exposed to more releases year-round or might differ from upwind schools.[16] Second, crosswind schools may receive some of the release, which would bias the estimates toward zero. I address both in Section 5. Third, identification relies on variation across facilities and school districts. Time-invariant differences across facilities are absorbed by the school × subject × grade fixed effects, but time-varying trends could still bias the estimates. In Section 5, I estimate specifications that compare schools around the same facility, with facility × year fixed effects, and with school district × year fixed effects.

Accidents happen in different years, so treatment is staggered. The recent difference-in-differences literature shows that the canonical TWFE estimator can be biased when treatment timing is staggered and effects differ across cohorts. I estimate the event study with the estimators of Sun and Abraham (2021) and Wing et al. (2024), which are robust to heterogeneous treatment effects under staggered timing, and show how their estimates compare with the main TWFE estimates.

The effects are intent-to-treat effects of being downwind on the accident day, as I do not observe the actual exposure of individual students. Accidents with off-site impacts, such as those leading to shelter-in-place orders, might also affect control schools, so my estimates are likely a lower bound on the magnitude of the total impact. Treated school-subject-grade cells with scores only after treatment are excluded; cells with only pre-period scores stay.

4 Results

I estimate the effect of being downwind of an airborne chemical accident with off-site impacts on school-level test scores, pooled across grades 3–8. I use two treatment definitions: airborne releases with off-site impacts (the main sample) and any accident with off-site impacts, which adds spills and runaway reactions. Figure 2 plots the event-study coefficients for math and reading. Pre-period coefficients are small and jointly insignificant for math and reading under both definitions.

[16] If a school is in the prevailing wind direction, it may experience disamenities (e.g., smell), which might cause some sorting, so downwind and non-downwind schools may not be comparable.

Exposure to an airborne accident lowers math scores by 0.040 SD in the first tested year after the accident. The post-period coefficients remain negative, and the average effect over event years 0 to +4 is −0.056 SD, which indicates that the decline is not a one-time effect. The average is less precisely estimated, partly because fewer schools are observed at longer horizons. In a balanced event window, the impact-year and average effects are −0.049 and −0.041 SD (Table 3). Including all accidents with off-site impacts gives very similar math estimates (Figure 3): −0.032 SD in the impact year and −0.055 SD on average.

Effects on reading are smaller and imprecise. Airborne accidents reduce reading scores by 0.018 SD in the impact year, and the average effect is −0.016 SD and not statistically significant. For all accidents with off-site impacts, the impact-year reading effect is close to zero (−0.008 SD). Table 2 reports the impact-year, average, and static estimates for both samples. Effects on reading are smaller and imprecise, a pattern also found for coal-fired power plant emissions, where English effects are about half the size of math effects (Duque and Gilraine, 2022).

Because schools are exposed in different years, two-way fixed-effects event studies can be biased when treatment effects vary across cohorts. Figure 4 re-estimates the main specification using the interaction-weighted estimator of Sun and Abraham (2021) and the stacked estimator of Wing et al. (2024). The three estimators produce nearly identical coefficients in both samples and both subjects, so I report TWFE estimates in the rest of the paper.

These effects reflect exposure to the release rather than other disruptions associated with an accident at a nearby facility. Because treated and control schools lie near the same facility, the estimates do not reflect where facilities are sited; they could, however, reflect the accident itself, such as emergency response or local disruption. Figure 6 repeats the estimation in equation (3) for accidents whose harm stayed on site.[17] If the main result is driven by exposure, the estimate for on-site accidents should be close to zero. To keep the placebo clean, I drop schools that were ever downwind of an accident with off-site impacts or environmental damage. The estimates are close to zero throughout the event window despite a larger sample of airborne on-site accidents. On-site spills and runaway reactions likewise show no effect. Accidents whose only off-site consequence was environmental damage are too few to estimate on their own; adding

[17] These accidents involve the same kinds of releases but far smaller ones (a median of 20 pounds released, against 450 for airborne accidents with off-site impacts).

them to the airborne sample leaves the results unchanged.

Figure 5 shows that the effect on math rises with the toxicity-weighted release rate. In the impact year, schools downwind of low-rate releases show no change, while those downwind of middle- and high-rate releases lose 0.051 and 0.063 SD. Average effects follow the same ordering (−0.029, −0.079, and −0.084 SD). Reading shows no clear gradient.

Accidents involving only flammable substances provide a useful contrast. EPA evaluates them by the risk of fire and explosion (overpressure and radiant heat) rather than by a toxic endpoint, so they do not enter $\mathcal{T}_a$. If the effect operates through inhaled chemicals rather than through the disruption of an emergency near the school, flammable-only accidents should have no effect. Estimated in the same regression, they do not show any impact on scores (Appendix Figure A4). By chemical, the effect appears for ammonia and other toxic substances but not for flammables, although pre-trends for ammonia are poor (Appendix Figure A4).

4.1 Heterogeneity

The main specification pools all accidents and all grades. Accidents differ, however, in which chemicals they release and how much, and they reach different student populations. Section 4 showed that the effect is concentrated in more toxic releases; this section asks which students are affected and summarizes how the effect varies with other accident characteristics. Throughout, I report the impact-year effect, the most precisely estimated quantity in Section 4, alongside the average post-period effect, and I test whether the groups' average effects are equal. Because each split divides a sample of 126 treated facility clusters, these tests have limited power, and I treat the patterns as suggestive.

I next re-estimate equation (3) separately for elementary grades (3–5) and middle-school grades (6–8), which are typically taught in different schools. Figure 7 shows that the math effect is larger in middle-school grades: −0.055 SD in the impact year and −0.079 SD on average, against −0.026 and −0.035 SD in grades 3–5. The difference is not statistically significant. Reading follows the same pattern (−0.031 versus −0.006 SD on impact). By single grade, grade 6 shows the largest effects in both subjects (Appendix Figure A1).

EDFacts also reports proficiency separately by sex and race. Figure 8 shows no difference between female and male students in math, while for race, the math effect is concentrated

among non-white students: −0.045 SD on impact and −0.072 SD on average, against −0.025 and +0.003 SD among White students.[18]

One channel through which accidents could affect test scores is lost instructional time, if schools close or shelter in place during the release. Two patterns suggest that this channel does not drive the results. First, accidents during school hours do not drive the impact-year effect: the drop in math scores is smaller for accidents during school hours than for accidents outside school hours or during the summer break, when no instruction can be lost (Appendix Figure A2). Second, when an accident triggers a shelter-in-place order, which may close schools, local officials typically apply it to a radius around the facility, often between half a mile and two miles. Upwind and downwind schools within that radius lose instructional time similarly, and 86.5 percent of treated schools lie more than 3 km (about two miles) from the accident that sets their treatment, a share nearly identical to that of control schools (86.0 percent). Together with the concentration of the effects in medium- and high-toxicity releases (Section 4), these patterns point to health effects of exposure rather than missed school days as the more likely channel. This interpretation is consistent with Goodman (2014), who finds that student absences, not school closures, reduce achievement.

5 Robustness

To interpret the estimates as the causal effect of airborne accidents with off-site impacts, treated and control schools must be comparable over time, and the wind direction on the accident day must be uncorrelated with underlying trends in school outcomes. I consider a narrower wind direction definition, the prevailing wind direction to rule out chronic exposures, and different test data specifications. This section also examines whether the composition of schools changes after an accident, to rule out possible sorting across schools in the estimation sample.

The exposure assignment uses a 45-degree cone around the downwind direction. To test how sensitive the results are to this choice, I narrow the cone to 22.5 degrees and, because a plume can reach some crosswind schools, restrict the comparison group to upwind schools

[18] EDFacts suppresses small subgroups, and White students are reported more consistently than other groups. I measure non-white students as all test takers minus White test takers.

only (Figure A3). The effect is concentrated within the narrower cone, and with upwind-only controls, the impact-year effect is similar, but it fades after the second year, so the average is smaller and not significant. Dropping ambiguous schools, which were upwind of one accident but downwind of another, gives similar results.

A further concern is that schools downwind on the accident day might be downwind most of the time. Because the wind carries odors and routine emissions toward them, disadvantaged communities may have sorted into prevailing-downwind neighborhoods, and if accidents happen in the prevailing downwind direction, part of the impact might come from this channel. School fixed effects absorb any permanent difference in who attends prevailing-downwind schools; sorting is a threat only if it creates different trends, for example if neighborhoods on the usual downwind side were declining. The left panel of Figure 9 shows, for each school, the share of days in a year that it lies inside the 45-degree downwind cone of its facility. Treated schools are more often downwind over the year than controls (33 versus 24 percent of days on average). I ask whether the effect appears among schools that are not on their facility's usual downwind side (an annual share below 25 percent, the share expected under random wind), and I add annual downwind-share bins interacted with year, which compare schools in a similar chronic position. Among treated schools that are not usually downwind, the impact-year effect on math is -0.053 SD and significant, at least as large as among prevailing-downwind schools (Figure 9, right panel). This suggests that the findings are not driven by chronically downwind schools.

Accident facilities differ: one may be a chemical plant, another a food-processing plant that stores ammonia. The neighborhoods around them may differ in ways that school fixed effects absorb but that could evolve differently over time. I add facility-group-by-year fixed effects, which compare downwind and non-downwind schools around the same facilities in the same year. School districts may also adopt policies or change spending in ways that create different trends across schools, so I separately add district-by-year fixed effects. Figure 10 shows both specifications. In both specifications, the math effect is similar to the main estimate, while with district-by-year fixed effects the reading effect becomes larger in magnitude and significant on average.

Because cohorts treated early contribute the late event years and cohorts treated late

contribute the early ones, I also estimate the event study on cohorts 2013–2016, which are observed in every event year from three years before to three years after the accident, so that the path is not driven by changing cohort composition. The impact-year effect is similar to the main estimate, and the average is slightly smaller. In addition, Guignet et al. (2023b) show that accidents reduce property values in a circle around the facility, which may create sorting. While I expect treated and control groups to experience similar property value drops, as they are within the same distance, I test whether school enrollment, the number tested, or schools' racial composition changes around the accidents. Figure 11 shows no drop in the number tested; enrollment drops, but not significantly; and the non-white share of test takers trends down, which would, if anything, raise scores rather than lower them. School composition therefore does not change differentially for treated schools compared with control schools.

EDFacts reports ranges rather than exact values in its public files. In Appendix Table A1, I re-estimate the main specification using the lower and upper bounds of each range and a standardized score whose standard deviation is corrected for the ranges; the impact-year and average effects are in line with the main findings.

6 Conclusion

This paper estimates the effect of chemical accidents on student learning. I use accident reports of airborne chemical accidents with off-site impacts between 2009 and 2018 and link them to nearby schools.

Using quasi-exogenous variation in wind direction on the day of the accident, I compare schools downwind of a release with schools at a similar distance from the same facility that were not downwind. Math scores fall by 0.040 school-level SD in the first test year after an accident, while the average effect over five years is −0.056 SD. Reading scores fall by less, and the effect fades over time and is less stable across specifications. Schools downwind of accidents that caused on-site harm but no reported off-site impact show no decline.

These effects are large for a single event. In student-level SD, the first-year math effect is comparable to the effect of a TRI facility opening within one mile of a school (Persico and Venator, 2021) and about half the effect of attending a school persistently downwind of a highway (Heissel et al., 2022), a chronic exposure. The effects are concentrated in accidents

with the highest toxicity-weighted release rates, which suggests that what was released matters, not only how long exposure lasts. Because test scores predict adult earnings, these losses may carry into the labor market. The persistent math decline is concentrated among non-white students: within the same schools, White students' scores return to baseline after the first year, while non-white students' scores remain lower. Because RMP facilities are disproportionately located near Black and Hispanic communities (U.S. EPA, 2024b), chemical accidents may also widen racial gaps in achievement.

EPA's 2024 Safer Communities by Chemical Accident Prevention rule strengthened accident prevention and expanded community access to facility hazard information, noting that about 131 million people live within three miles of an RMP facility. A February 2026 proposal would rescind much of the rule, including the requirement that facilities provide hazard information to nearby residents on request (U.S. EPA, 2026a). The rule's regulatory impact analysis monetizes fatalities, injuries, hospitalizations, evacuations, sheltering in place, and property damage, but not effects on children's learning. To the extent that these estimates generalize, this omission understates the benefits of accident prevention.

This paper uses publicly available school-level test scores. These data cannot separate lost learning from changes in which students take the test, nor distinguish among channels such as health, absences, and school disruption. Student-level administrative data would allow both, and would also make it possible to test whether information about nearby facilities helps families avoid exposure.

7 Figures and Tables

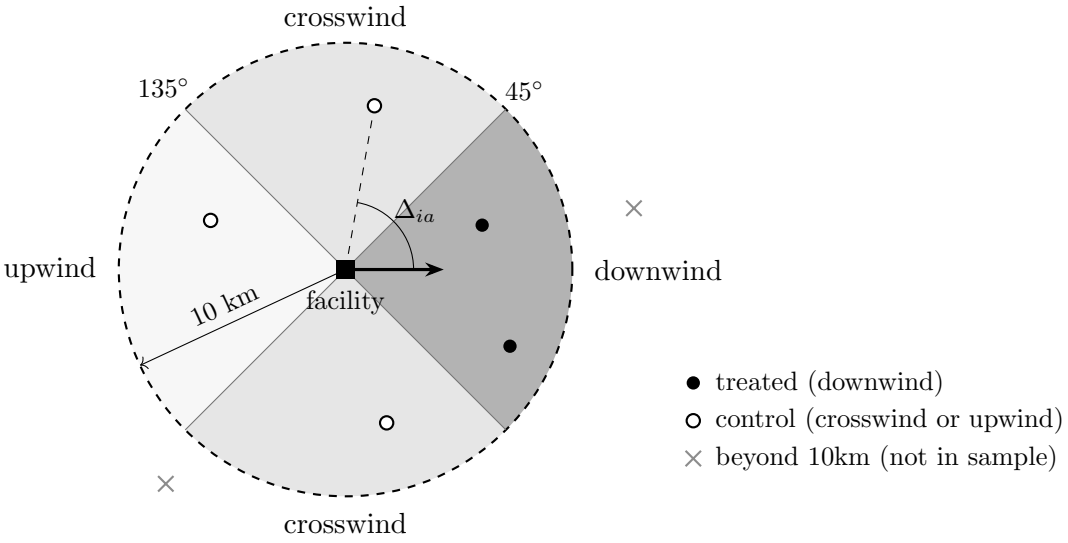


Figure 1: Classifying schools by wind direction around an accident

Note: The arrow from the facility points in the downwind direction, defined by the vector-mean 10 m wind on the local calendar day of the accident. Δ_{ia} is the angle between the downwind direction and the bearing from the facility to school i. A bearing is a compass direction in degrees, measured clockwise from north: 0° is north, 90° east, 180° south, 270° west. Schools within 10 km are downwind if $\Delta_{ia} \leq 45°$, crosswind if $45° < \Delta_{ia} \leq 135°$ on either side, and upwind if $\Delta_{ia} > 135°$. A school is treated if it is downwind of at least one accident; schools within 10 km of an accident but never downwind of one are controls.

Table 1: Pre-accident characteristics of treated and control schools

	Treated (1)	Control (2)
Panel A. Students and test scores (EDFacts, spring 2010)		
Test score (school SD)	-0.200 [0.926]	-0.276 [1.012]
Log(students tested), math	5.608 [0.745]	5.555 [0.790]
Non-white share of students tested	0.636 [0.335]	0.615 [0.336]
Panel B. School characteristics (CCD, school years 2008-09 and 2009-10)		
Free or reduced-price lunch share	0.612 [0.270]	0.609 [0.285]
Log(enrollment)	6.192 [0.540]	6.153 [0.575]
City	0.441	0.485
Suburb	0.428	0.401
Town	0.058	0.050
Rural	0.073	0.064
Charter school	0.050	0.098
Elementary school	0.738	0.732
Middle school	0.217	0.192
High school	0.013	0.036
Schools	1,182	3,659

Note: Means, with standard deviations of continuous characteristics in brackets. Treated schools are the identifying treated schools of the main sample (downwind of an airborne accident with off-site impacts and with a post-accident test in the estimation sample); control schools are the controls of the same sample. All characteristics predate the first accident of any treated school (April 2010): test outcomes are from spring 2010, and the earliest first exposed test is spring 2011. Non-white share is one minus the White share of students tested. Free or reduced-price lunch share and enrollment average the CCD counts of school years 2008-09 and 2009-10 (taken each October), before the Community Eligibility Provision; zero counts are treated as missing. Locale, charter status and school level are each school's modal CCD value; school-level shares do not sum to one because the remaining schools have other grade configurations. Schools missing a characteristic are excluded from that row.

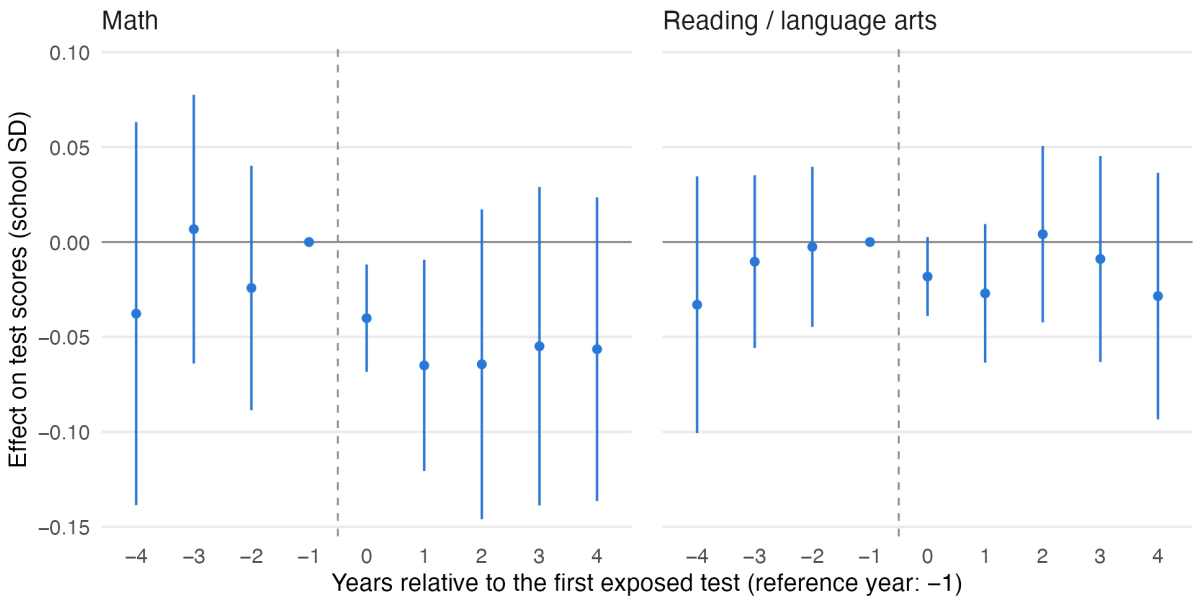


Figure 2: Test scores after airborne accidents with off-site impacts

Note: This figure plots event study coefficients for Equation 3. Event time k runs from -4 to $+4$ test years relative to the accident, where $k = 0$ is the first test after the accident; $k = -1$ is the omitted reference period. The dependent variable is standardized school-level test score, observed from 2009–2019 for math and reading. The treatment variable is equal to 1 for schools within 10 km of an accident facility and downwind on the accident day; the estimating sample is schools within 10 km of accident facilities. Regression controls include school × subject × grade FE; and state × subject × grade × year FE; see text for details.

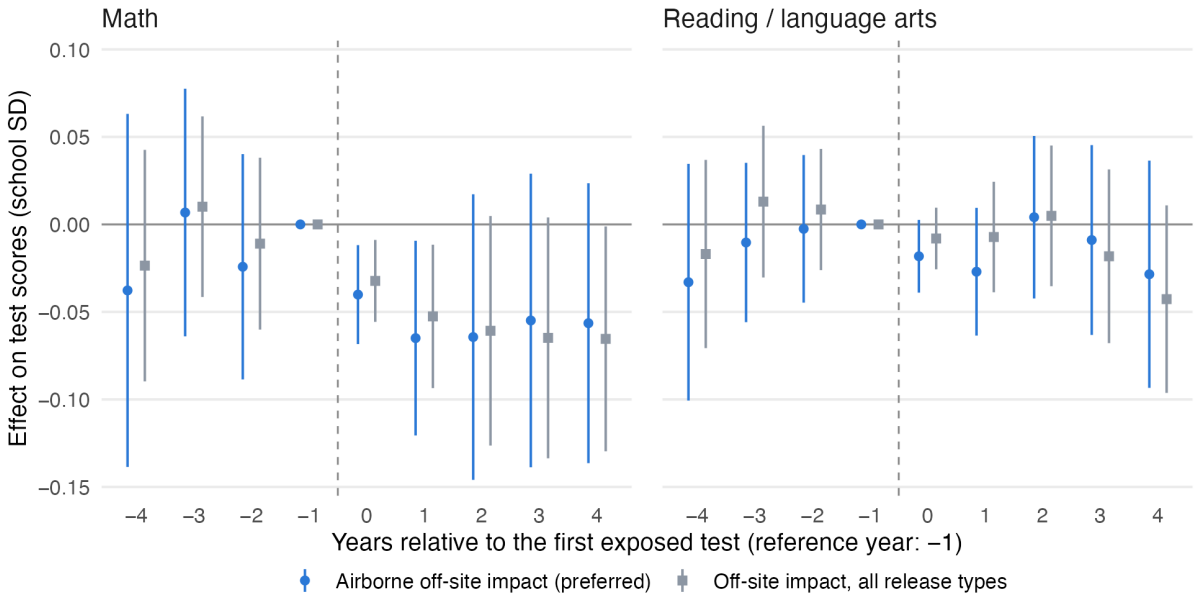


Figure 3: Test scores after airborne off-site accidents and all off-site-impact accidents

Note: This figure plots event study coefficients for Equation 3. The dependent variable is standardized school-level test score, observed from 2009–2019 for math and reading. The treatment variable is equal to 1 for schools within 10 km of an accident facility and downwind on the accident day; the estimating sample is schools within 10 km of accident facilities. Regression controls include school × subject × grade FE; and state × subject × grade × year FE; see text for details. The two series differ only in the accident set: airborne accidents with off-site impacts (preferred sample) and all accidents with off-site impacts regardless of release type.

Table 2: Effect of downwind exposure to chemical accidents on test scores

	Airborne offsite impact (preferred)			Offsite impact, all release types		
	Math	RLA	Both	Math	RLA	Both
Immediate effect (k = 0)	-0.040***	-0.018*	-0.029**	-0.032***	-0.008	-0.020**
	(0.014)	(0.011)	(0.011)	(0.012)	(0.009)	(0.009)
Average effect (k = 0 to +4)	-0.056*	-0.016	-0.036	-0.055**	-0.014	-0.035*
	(0.031)	(0.019)	(0.023)	(0.024)	(0.016)	(0.019)
Treated schools	1,180	1,182	1,182	1,550	1,552	1,553
Control schools	3,653	3,647	3,659	4,989	4,983	4,997
Observations	143,536	144,219	287,755	192,010	192,902	384,912

Note: TWFE event study weighted by the number tested; outcome z (school SD); fixed effects school × subject × grade and state × subject × year × grade; standard errors clustered by facility group in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. k = 0 = first test after the accident; average = event years 0 to +4, equal weights. Treated schools = identifying (a post-treatment row in the estimation sample). Observations = school × subject × grade × year.

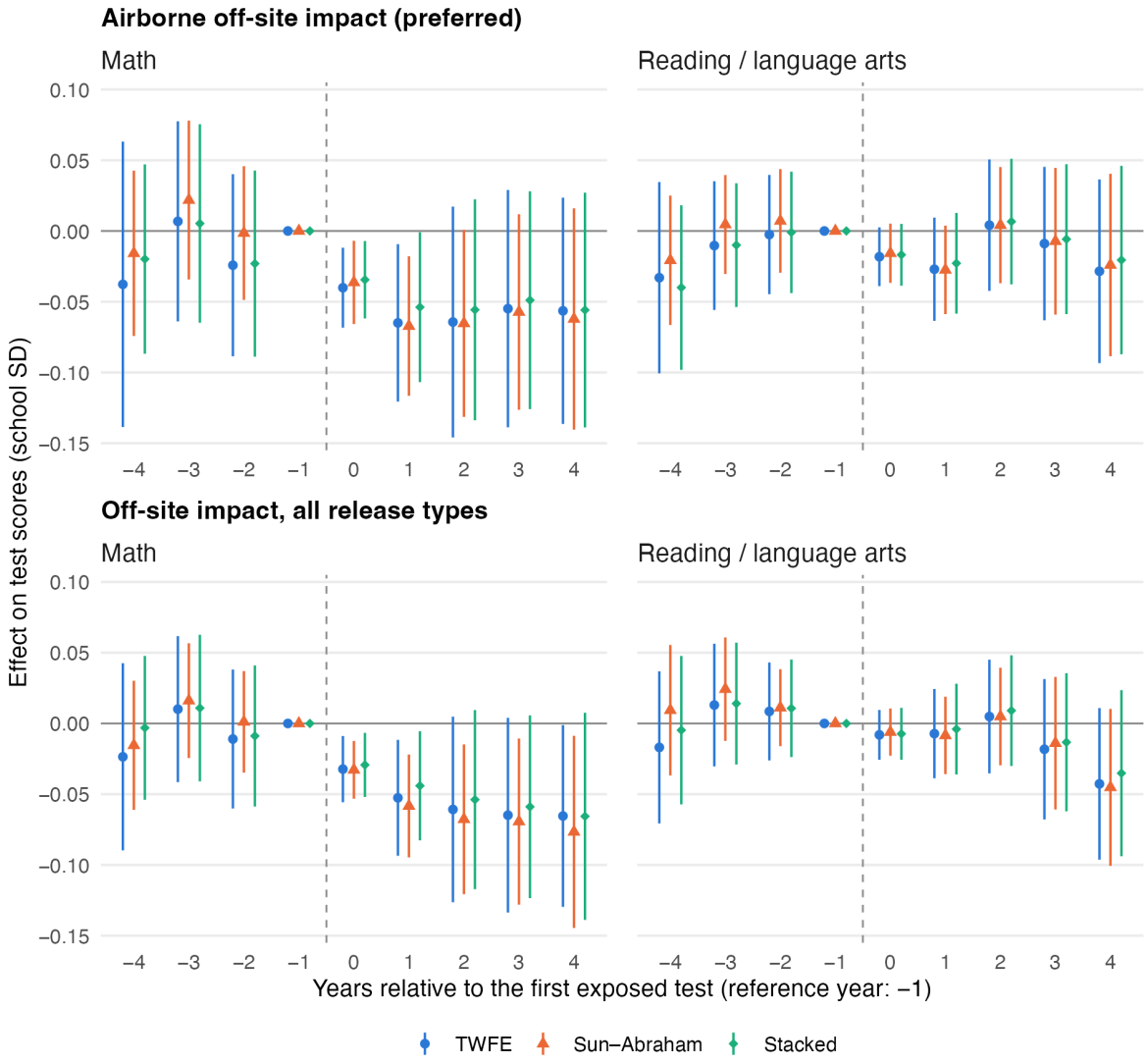


Figure 4: Test scores after airborne off-site accidents: alternative staggered-treatment estimators

Note: This figure plots event study coefficients for each estimator. The dependent variable is standardized school-level test score, observed from 2009–2019 for math and reading. The treatment variable is equal to 1 for schools within 10 km of an accident facility and downwind on the accident day; the estimating sample is schools within 10 km of accident facilities. Regression controls include school × subject × grade FE; and state × subject × grade × year FE; see text for details.

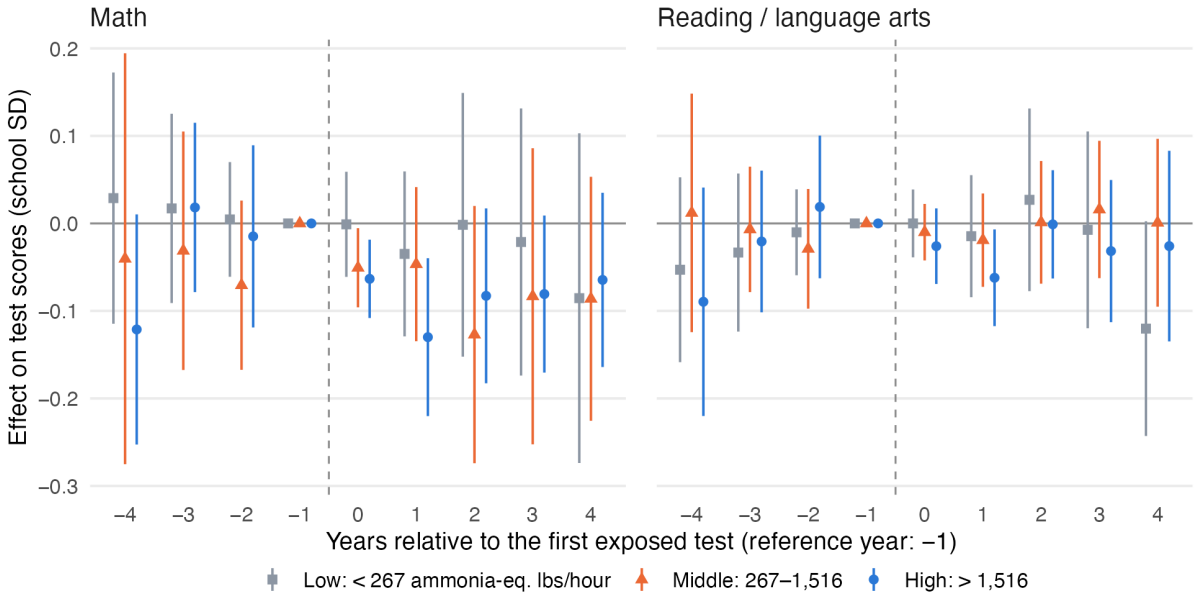


Figure 5: Test scores after airborne off-site accidents, by toxicity-weighted release rate

Note: This figure plots event study coefficients for Equation 3, with treatment interacted with three bins of the accident's toxicity-weighted release rate and with cut-offs defined by the terciles of the release rates of accidents: low (< 267 ammonia-equivalent lbs/hour), middle (267–1,516), and high (> 1,516). The toxicity-weighted release rate is the quantity released per hour multiplied by the ratio of the toxic endpoint of anhydrous ammonia to the chemical's toxic endpoint, as ammonia-related accidents are the most prevalent, so one unit carries the toxic hazard of one pound of ammonia released per hour. The dependent variable is standardized school-level test score, observed from 2009–2019. The treatment variable is equal to 1 for schools within 10 km of an accident facility and downwind on the accident day; the estimating sample is schools within 10 km of accident facilities. Regression controls include school × subject × grade FE and state × subject × grade × year FE. Vertical bars show 95% confidence intervals with standard errors clustered by facility group; see text for details.

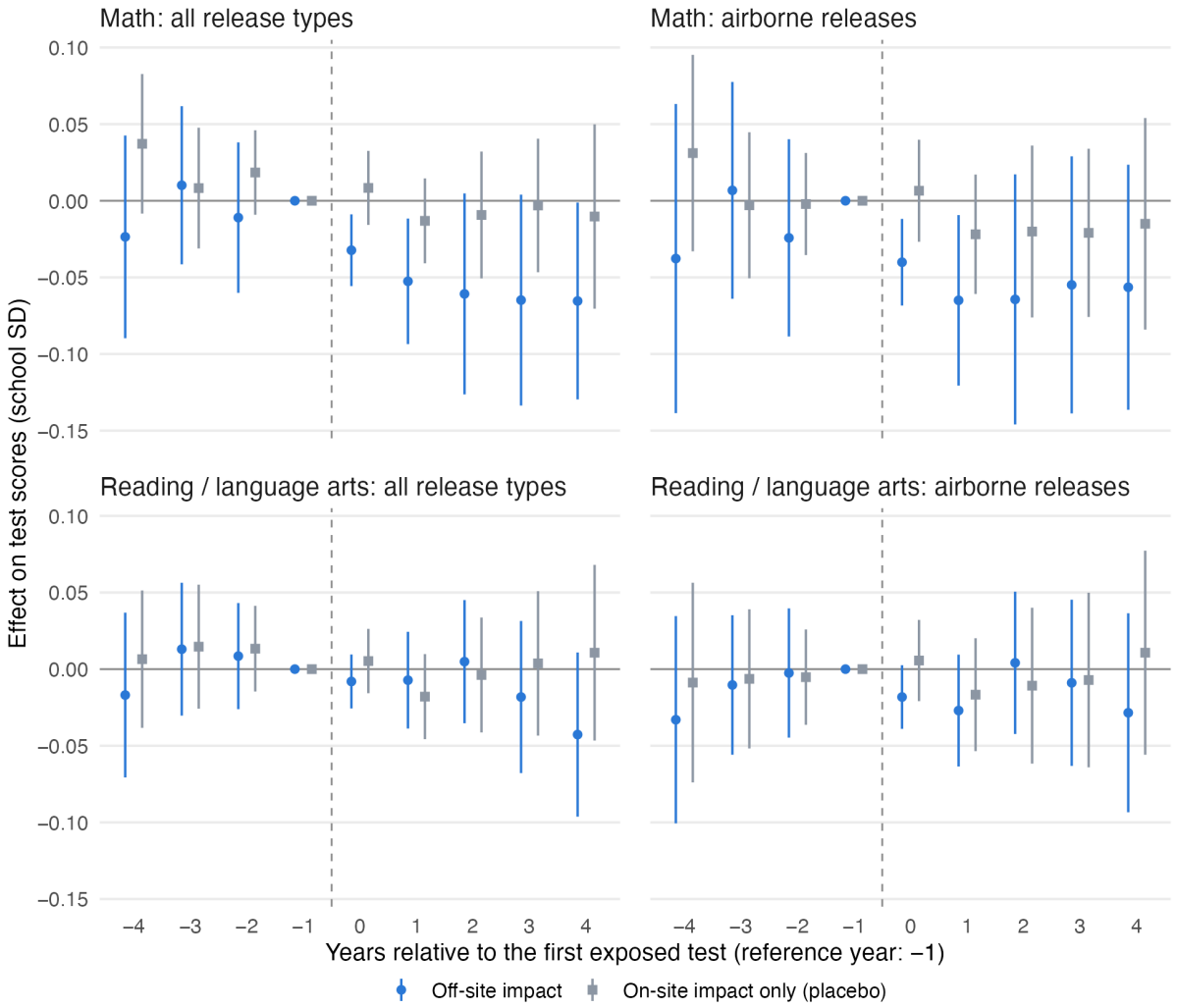


Figure 6: Test scores after off-site and on-site accidents

Note: This figure plots event study coefficients for Equation 3. The left panel shows math and the right panel reading/language arts. The dependent variable is standardized school-level test score, observed from 2009–2019. The treatment variable is equal to 1 for schools within 10 km of an accident facility and downwind on the accident day, separately for off-site and on-site accidents and for airborne and non-airborne releases; the estimating sample is schools within 10 km of accident facilities. Regression controls include school × subject × grade FE and state × subject × grade × year FE. Vertical bars show 95% confidence intervals with standard errors clustered by facility group; see text for details.

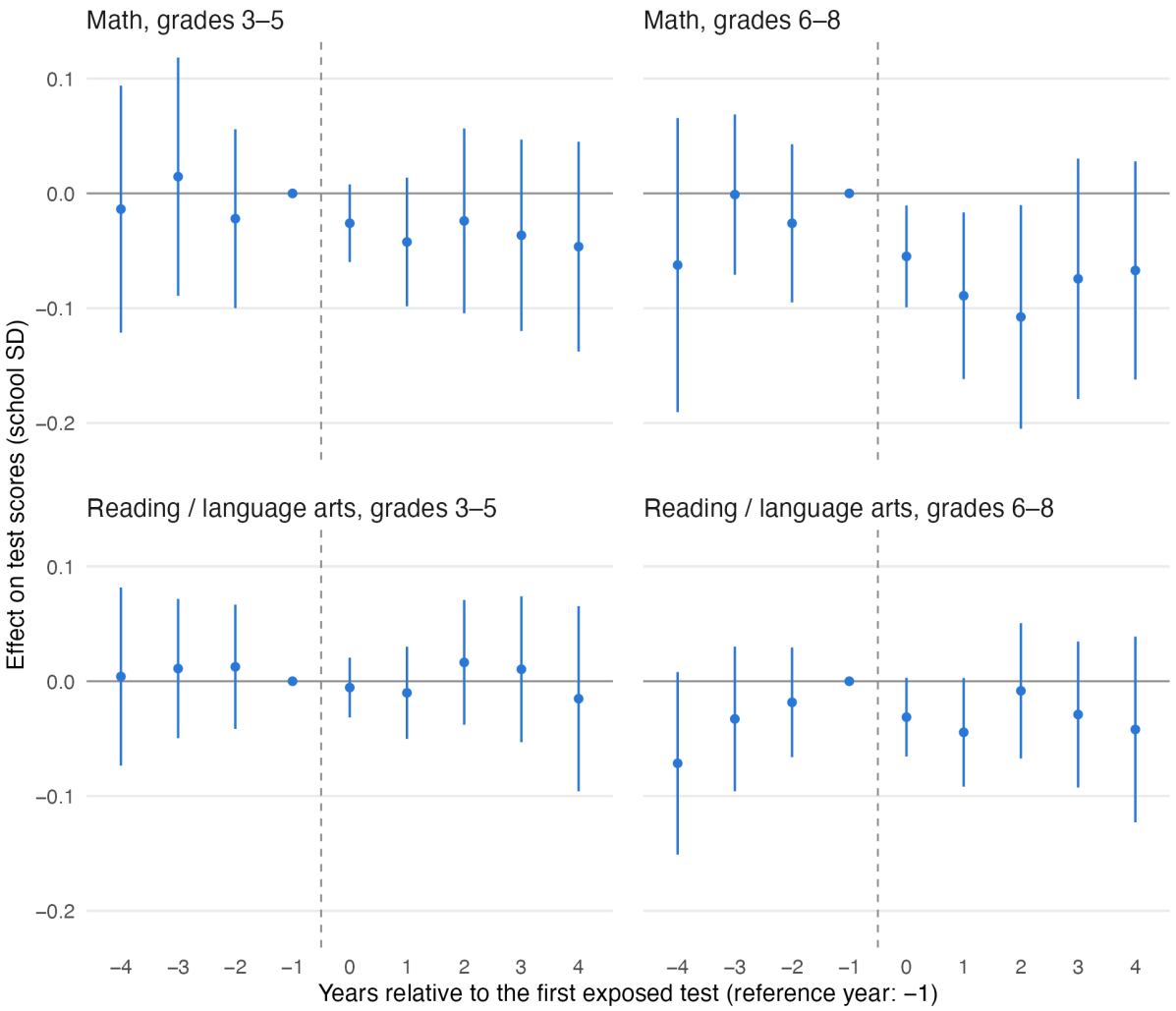


Figure 7: Test scores after airborne off-site accidents: grade level

Note: This figure plots event study coefficients for Equation 3 for each grade level instead of pooled regression.

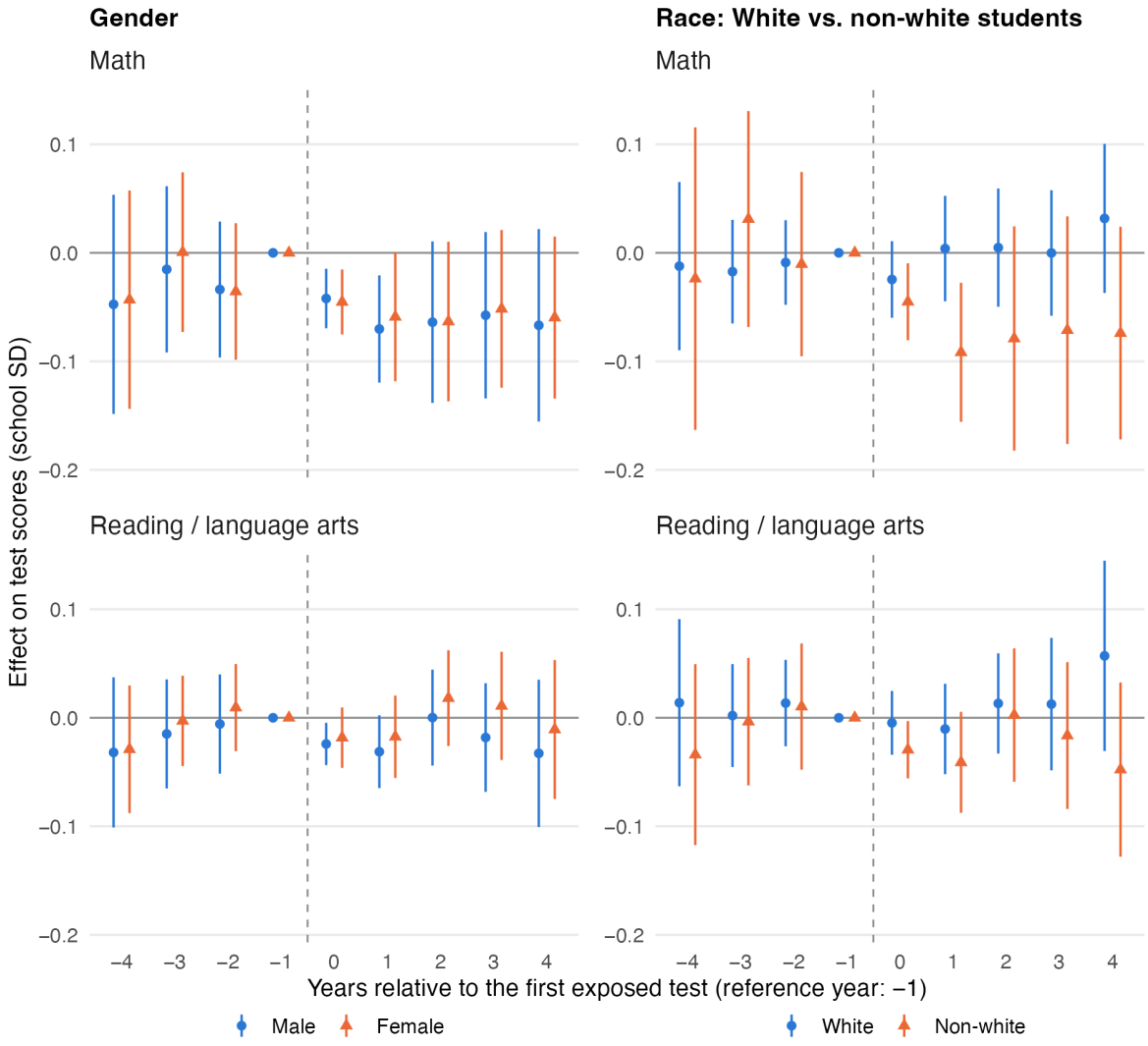


Figure 8: Test scores after airborne off-site accidents: heterogeneity by subpopulation

Note: This figure plots event study coefficients for Equation 3 for each subgroup separately, with the same estimation. Because schools are more likely to report counts for White students, I subtract White students from all students to obtain non-white counts.

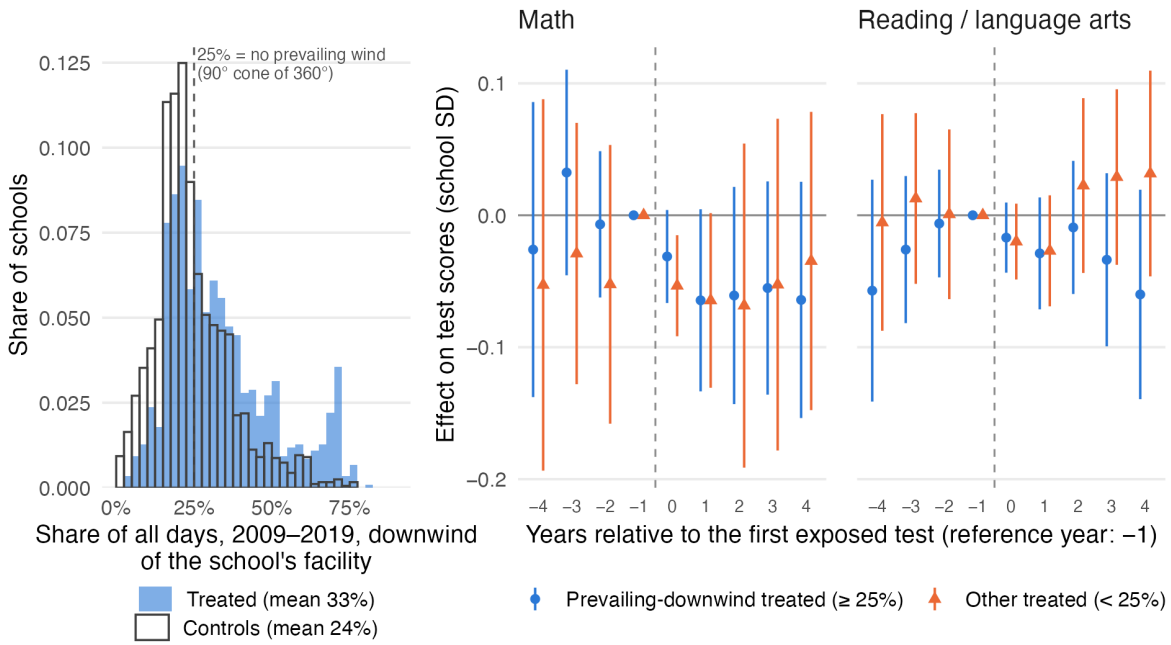


Figure 9: Test scores after airborne off-site accidents: prevailing wind direction

Note: The left panel shows, for each school, the share of all days in 2009–2019 on which the wind blew from its facility toward the school (the annual downwind share), for treated schools (mean 33%, $n = 1{,}182$) and control schools (mean 24%, $n = 3{,}659$). The dashed line at 25% marks no prevailing wind. The right panel plots event study coefficients for Equation 3, splitting treated schools into those downwind of their facility on at least 25% of days and those downwind on less than 25% of days. Both groups are estimated in a single regression against the same 3,529 control schools.

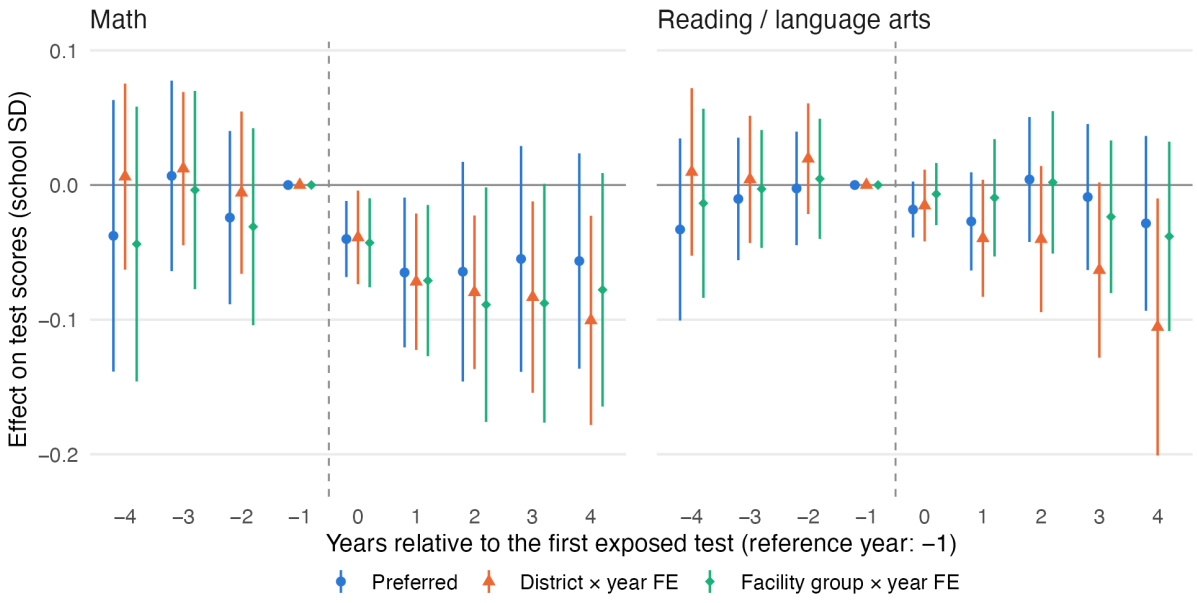


Figure 10: Test scores after airborne off-site accidents: different specifications

Note: This figure plots event study coefficients for Equation 3 under two additional fixed-effect specifications, each added to the main fixed effects. Facility-group × year fixed effects compare downwind and non-downwind schools around the same facility group in the same year; district × year fixed effects absorb time-varying shocks common to all schools in a school district.

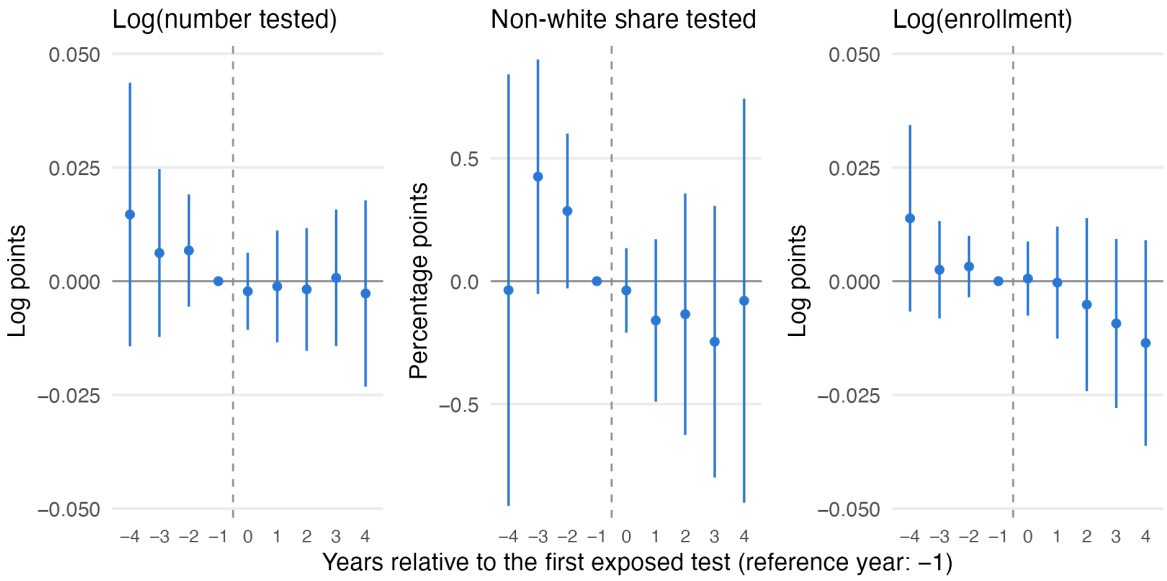


Figure 11: School composition after airborne off-site accidents

Note: This figure plots event study coefficients for Equation 3 for school composition dependent variables. Each panel re-estimates the event study with a measure of the school's student body as the outcome; flat post-accident estimates indicate that accidents did not change who was enrolled or tested. Log(number tested) and non-white share: school × grade × year cells (math), with school × grade and state × year × grade fixed effects, weighted by the number tested. Log(enrollment): school × year (CCD), with school and state × year fixed effects, unweighted; k = 0 is the first October enrollment count after the accident. Standard errors clustered by facility group.

Table 3: Robustness to sample and specification choices

	Math		Reading / language arts	
	$k = 0$	Average	$k = 0$	Average
Preferred	-0.040***	-0.056*	-0.018*	-0.016
	(0.014)	(0.031)	(0.011)	(0.019)
Panel A. Which schools are compared				
Without charter and high-level schools	-0.042***	-0.062*	-0.019*	-0.017
	(0.014)	(0.031)	(0.011)	(0.020)
Schools open through 2019	-0.038**	-0.054*	-0.017	-0.016
	(0.015)	(0.031)	(0.011)	(0.019)
Clean controls (never downwind of any accident)	-0.045***	-0.069**	-0.022**	-0.030
	(0.014)	(0.031)	(0.011)	(0.022)
Drop middle-school controls with a treated feeder school	-0.041***	-0.057*	-0.018*	-0.017
	(0.015)	(0.031)	(0.011)	(0.019)
Panel B. Defining treatment				
Cut at a second exposure	-0.040***	-0.057*	-0.018*	-0.014
	(0.014)	(0.031)	(0.011)	(0.019)
Drop treated middle schools with an earlier-treated feeder	-0.040***	-0.057*	-0.018*	-0.016
	(0.014)	(0.031)	(0.011)	(0.019)
Moved schools included	-0.039***	-0.054*	-0.020*	-0.016
	(0.014)	(0.030)	(0.010)	(0.019)
School year from June 1	-0.040***	-0.056*	-0.018*	-0.016
	(0.014)	(0.031)	(0.011)	(0.019)
Panel C. Balanced samples				
Balanced schools (cells observed all 10 years)	-0.034*	-0.071*	-0.017	-0.029
	(0.019)	(0.039)	(0.013)	(0.022)
Balanced event window (cohorts 2013-2016, k -3..+3)	-0.049**	-0.041	-0.029**	-0.005
	(0.019)	(0.030)	(0.014)	(0.020)
Panel D. Additional controls				
Composition controls	-0.045***	-0.060**	-0.024**	-0.020
	(0.014)	(0.030)	(0.010)	(0.019)

Note: TWFE event study weighted by the number tested; outcome z (school SD); fixed effects school × subject × grade and state × subject × year × grade; standard errors clustered by facility group in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. k = 0 = first test after the accident; average = event years 0 to +4, equal weights. Balanced event window: average over k = 0 to +3.

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A Supplemental Appendices

Supplemental Figures and Tables

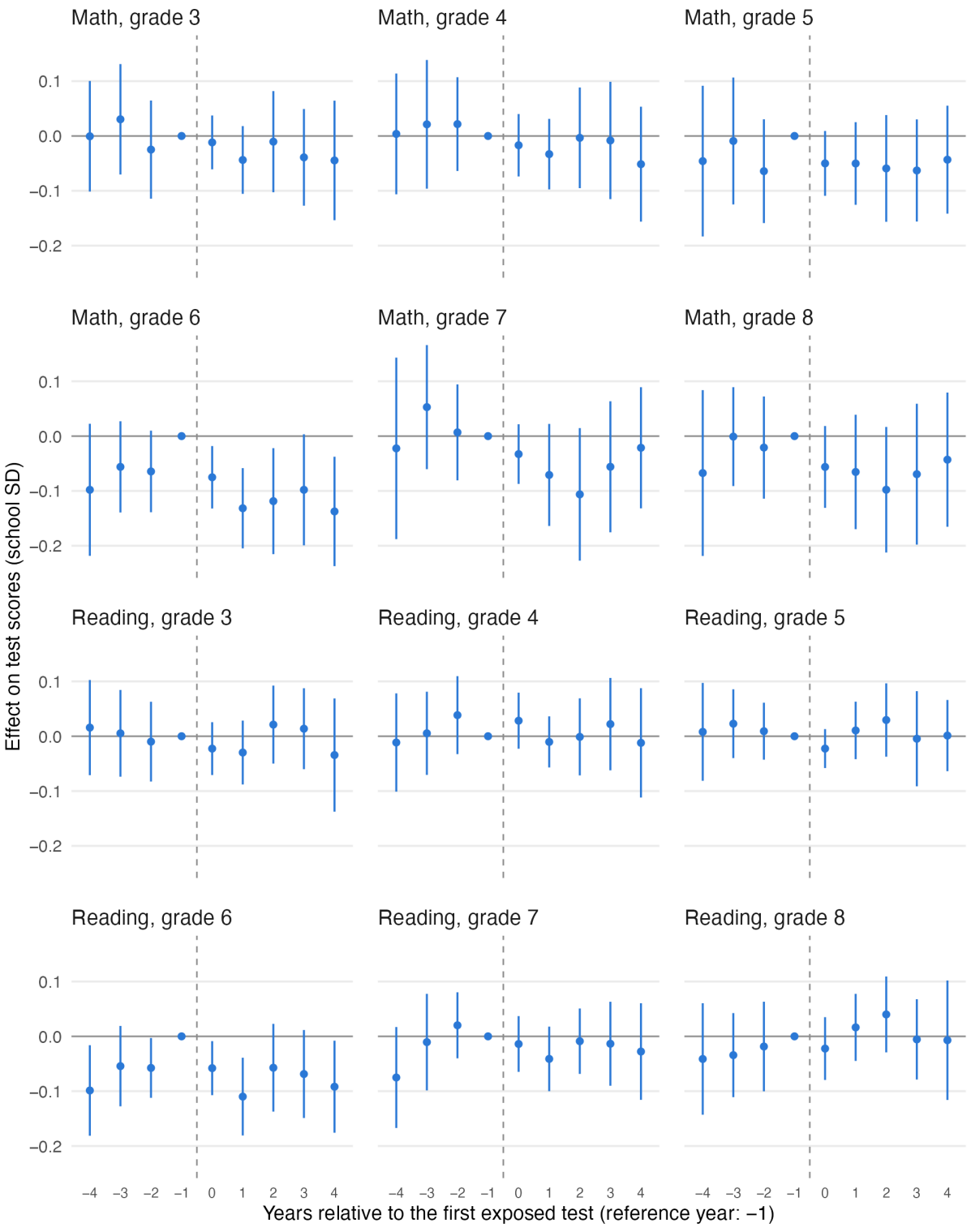


Figure A1: Test scores after airborne off-site accidents by grade

Note: This figure plots event study coefficients for Equation 3 for each grade level instead of pooled regression.

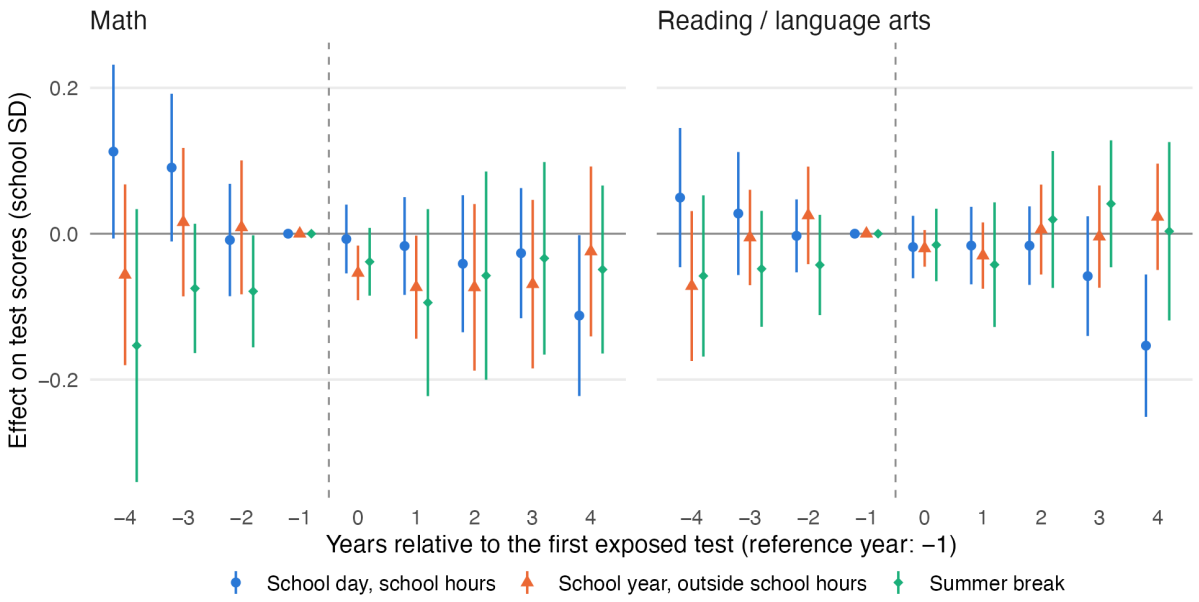


Figure A2: Test scores after airborne off-site accidents by accident timing

Note:

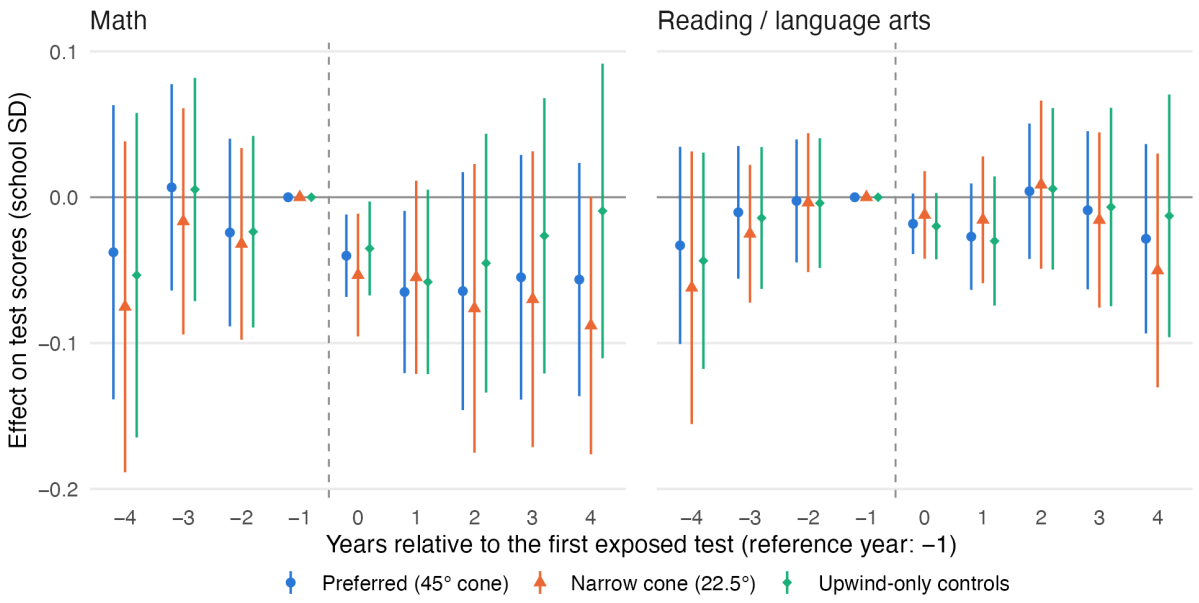


Figure A3: Test scores after airborne off-site accidents by wind specification

Note:

Table A1: Robustness to Test Changes and Measurement

	Math		Reading / language arts	
	$k = 0$	Average	$k = 0$	Average
Preferred	-0.040***	-0.056*	-0.018*	-0.016
	(0.014)	(0.031)	(0.011)	(0.019)
Panel A. Data coverage and reported ranges				
State-years with $\geq$ 80% of the state's schools reporting	-0.040***	-0.056*	-0.018*	-0.016
	(0.014)	(0.031)	(0.011)	(0.019)
Score at the bottom of the reported range	-0.040***	-0.058*	-0.017	-0.014
	(0.015)	(0.031)	(0.010)	(0.019)
Score at the top of the reported range	-0.040***	-0.055*	-0.019*	-0.017
	(0.014)	(0.030)	(0.011)	(0.019)
SD corrected for the reported ranges	-0.041***	-0.056*	-0.018*	-0.015
	(0.015)	(0.032)	(0.011)	(0.019)
Panel B. Other outcome scales				
Probit of the share proficient (student SD)	-0.017**	-0.024	-0.007	0.001
	(0.008)	(0.016)	(0.006)	(0.010)
Percentile rank (percentile points)	-1.05**	-0.95	-0.33	0.07
	(0.42)	(0.82)	(0.32)	(0.53)
Percent proficient (percentage points)	-0.70***	-1.14**	-0.34	-0.30
	(0.26)	(0.55)	(0.21)	(0.36)

TWFE event study weighted by the number tested; outcome z (school SD); fixed effects school $\times$ subject $\times$ grade and state $\times$ subject $\times$ year $\times$ grade; standard errors clustered by facility group in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. k = 0 = first test after the accident; average = event years 0 to +4, equal weights. Panel A: z from the bottom or top of each EDFacts range instead of its midpoint, or with the standard deviation net of the noise the ranges add. Panel B: probit = inverse normal of the share proficient; percentile rank among the balanced schools of the state $\times$ subject $\times$ grade $\times$ year. Treated schools (math): 1,180 in the preferred row; without break cohorts 786; without the 2015 cohort in states with a 2015 test change 1,000; percentile rank 1,035; within 1 percent of the preferred row in the others.

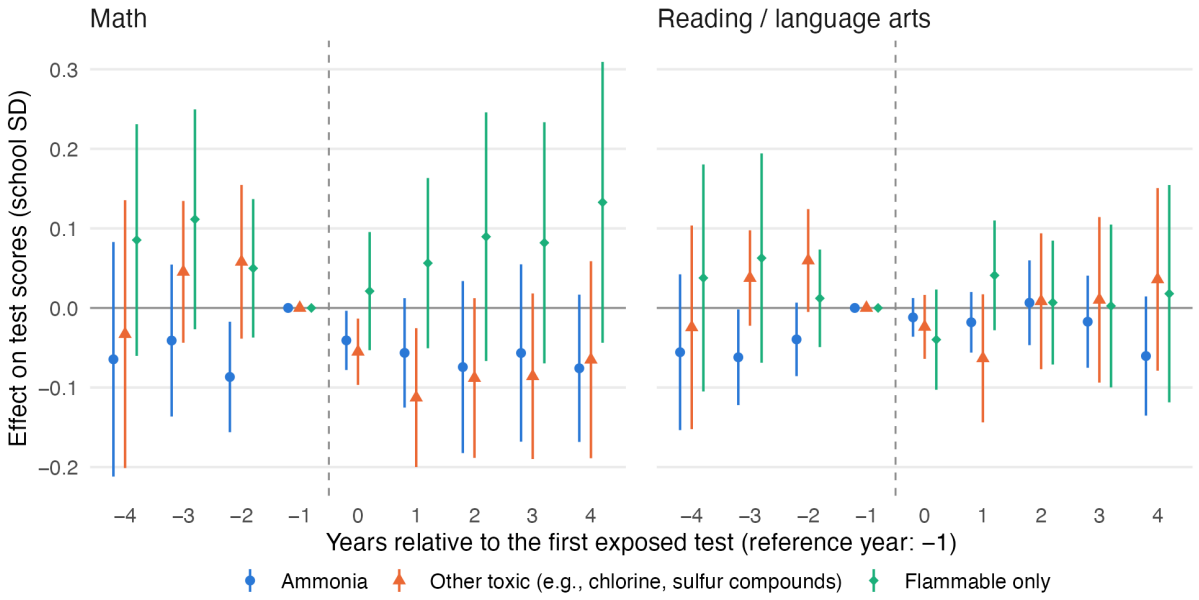


Figure A4: Test scores after airborne off-site accidents by chemical type

Note: This figure plots event study coefficients for Equation 3, with treatment interacted with the chemical released: ammonia, other toxic substances (e.g., chlorine, sulfur compounds), and flammable substances only, which have no toxic endpoint. All groups are estimated in a single regression against the same 3,529 control schools. Treated schools (math): ammonia 669 (88 clusters), other toxic 367 (25 clusters), flammable only 144 (26 clusters). The dependent variable is standardized school-level test score, observed from 2009–2019. Regression controls include school × subject × grade FE and state × subject × grade × year FE. Vertical bars show 95% confidence intervals with standard errors clustered by facility group.

B Data Appendix

B.1 Chemical Accidents

The accident data are compiled from EPA's Risk Management Plan (RMP) records, obtained through Freedom of Information Act (FOIA) requests by the Data Liberation Project. RMP records are self-reported: under Section 112(r) of the Clean Air Act, facilities that hold more than a threshold quantity of a regulated toxic or flammable substance must submit them. The records include detailed information about each facility, including its latitude and longitude and a unique EPA facility identifier, which allows me to link the records to other EPA data sources. Although RMP reports the facility's location, this is not the location of the accident (U.S. EPA, 2026b). The RMP reference point may be the plant gate, the center of the facility, or a storage tank, so I cross-check the coordinates reported in the filing that recorded each accident against the facility's location in EPA's Facility Registry Service (FRS), which the RMP data also include (Data Liberation Project, 2025). If the two locations differ by more than 5 km, I drop the accident from the sample; accidents without an FRS location are kept, and in all cases I use the RMP location because it is the contemporaneous measure.

Each facility must update its risk management plan at least once every five years, and an accident that meets the reporting criteria must be added to the plan's five-year accident history within six months of the release. An accident has off-site impacts if it causes known deaths, injuries, evacuations, sheltering in place, property damage, or environmental damage beyond the facility's property boundary or in areas of the site open to the public, and on-site impacts if it causes deaths, injuries, or significant property damage on site. Each accident can involve one or more chemicals and one or more release events, which are recorded for the accident as a whole rather than for each chemical. EPA defines a gas release as a release of the substance as a gas rather than vaporized from a liquid; a liquid spill/evaporation as a release in a liquid state with subsequent vaporization; a fire as combustion producing light, flames, and heat; an explosion as a rapid chemical reaction producing heat and expansion of gases; and an uncontrolled/runaway reaction as an uncontrolled chemical reaction that generates excessive heat, pressure, or harmful reaction products (U.S. EPA, 2009a). I classify an accident as airborne if it involves a gas release, a fire, or an explosion, and as spill/runaway if its only release

events are liquid spills or runaway reactions.

For each accident, the RMP reports on-site deaths and injuries; known off-site deaths, injuries (people hospitalized or given medical treatment), and the number of people evacuated or sheltered in place; and whether off-site responders were notified. Table A2 reports the number of accidents and facilities in each group between 2009 and 2018, along with the share of accidents by release type and by consequence. Off-site impact refers to off-site deaths, injuries, evacuations, sheltering in place, or property damage; on-site impact only refers to harm or damage inside the facility with no off-site consequence, including environmental damage. Release-type shares can sum to more than one because an accident can involve several release events. Off-site injuries are defined as people hospitalized or given medical treatment, and the evacuation share is the fraction of accidents in which at least one person was evacuated or sheltered in place.

Table A2: RMP accidents by consequence and release type

			Release type (share)				Consequences (share)		
	Accidents	Facilities	Gas	Fire	Explosion	Spill	On-site deaths or injuries	Off-site deaths or injuries	Off-site evacuation or shelter
Off-site impact, airborne	283	253	0.93	0.12	0.06	0.13	0.34	0.32	0.73
Off-site impact, spill / runaway	93	88	0.00	0.00	0.00	0.98	0.40	0.45	0.62
On-site impact only, airborne	645	467	0.86	0.25	0.07	0.09	0.71	0.00	0.00
On-site impact only, spill / runaway	221	178	0.00	0.00	0.00	1.00	0.86	0.00	0.00

Note: Accidents reported to EPA's Risk Management Program (RMP), 2009–2018. Airborne = a gas release, fire or explosion; spill / runaway = a spill or runaway reaction only. Consequence columns are shares of accidents in the row; see text for definitions.

B.2 School Location and Test Scores

The CCD provides school locations, including latitude and longitude coordinates, and school characteristics. I downloaded all school files from 2009 to 2019 and checked each school's location year by year; if there is a disagreement greater than 1 km, I drop the school. The CCD also identifies virtual schools; because a fully virtual school's CCD coordinates are an office while its students live all over the state, exposure cannot be assigned confidently. The Community Eligibility Provision (CEP) is a federal meal service option that allows high-poverty schools and school districts to serve breakfast and lunch at no cost to all enrolled students without collecting household applications. Once a school adopts it, its FRPL count stops reflecting student poverty. I only use it for school years 2008–09 and 2009–10, as a balance check.

Test scores: EDFacts school-level proficiency rates

EDFacts publishes test score data for every school, grade, subject, and year: the number of students who took the test and received a valid score, and the percent of them who scored at or above the level the state defines as proficient. To protect student privacy, individual students' scores are not reported, and EDFacts applies two further privacy protections. First, it suppresses every cell with 1–5 students tested and provides no information beyond a suppression flag. Second, it "blurs" the percent proficient of every other cell into a range whose width depends on how many students were tested: the fewer the students, the wider the range, and for cells with more than 300 students it reports whole numbers. Table A3 shows how the cells of the estimation sample are distributed across these ranges. The majority of cells are published in 5-point ranges, and only 2 percent are suppressed and dropped from the estimation sample.

Table A3: Distribution of EDFacts percent-proficient ranges in the estimation sample

			Share (%)	
Students tested	Published as	Example	Cells	Students tested
1–5	suppressed (dropped)	PS	2.1	0.1
6–15	two ranges: <50%, ≥50%	≥50%	1.9	0.2
16–30	20-point ranges	40–59%	6.4	1.6
31–60	10-point ranges	30–39%	26.5	12.8
61–300	5-point ranges	40–44%	59.1	68.6
over 300	whole numbers	47%	4.1	16.8

Note: Ranges as documented in EDFacts (2020). Shares are over the cells (school × grade × subject × year) of the 4,868 schools of the estimation sample, grades 3-8, spring 2010-2019: the 289,057 cells of the estimation sample and the 6,066 suppressed cells of the same schools, 28.7 million students tested in all. Suppressed cells have no score and are dropped from the estimation sample.

Because EDFacts publishes intervals rather than exact values, the published data carry their own measurement error. I treat each published value as an interval that contains the true percent proficient and construct the outcome in three steps.

Step 1: rounding. I read each published value as a range of rounded whole percents, so the true percent lies within half a point of it: 47% becomes $[46.5, 47.5]$, 40–44% becomes $[39.5, 44.5]$, and $\geq$ 95% becomes $[94.5, 100]$. The documentation does not say whether the ranges refer to rounded or unrounded percents, but only the rounded reading lets the ranges cover 0–100 without gaps: a school at 79.6% would otherwise fall in neither 60–79% nor $\geq$ 80%. If this reading were wrong, every interval would shift by half a point in the same direction, which the

standardization in Step 3 removes.

Step 2: whole students. With N students tested, the number of proficient students k is a whole number, so the percent proficient can only take the values $100k/N$. I keep the values of k that fall inside the interval from Step 1 and set

$$p^{\text{lo}} = \frac{100\,k_{\min}}{N}, \qquad p^{\text{hi}} = \frac{100\,k_{\max}}{N}. \tag{4}$$

This narrows the interval, most for small schools.

Step 3: one number per cell. My measure of percent proficient is the midpoint $p = (p^{\text{lo}} + p^{\text{hi}})/2$, which I standardize within each state, year, subject, and grade, using the mean and standard deviation of a balanced panel of schools (Jacqz, 2022). I keep both ends of the interval and show in the robustness section that the results do not depend on using the midpoint.